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Palladium catalysed, three-component cascade reaction for the efficient synthesis of diverse small molecules

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ABSTRACT

A palladium catalysed, three-component cascade reaction was employed for the regiospecific and stereoselective synthesis of novel substituted (*Z*)-2-butene 1,4-diamine analogues and substituted 4-aryloxy but-2-enyl amine analogues. Allenes were first synthesised from the corresponding alkynes using the Crabbé reaction and then reacted with a large range of commercially available aryl iodides and amines to generate a library of small molecules. The reaction is compatible with aryl iodides containing both electron-donating and electron-withdrawing groups. Amines of different sizes can also be utilised with no noticeable effect on the yield, regiospecificity or stereoselectivity.

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Introduction

One-pot multicomponent reactions (MCRs) are a powerful method for synthesizing complex molecular scaffolds in a single step [1–3]. Advantages in using MCRs over a linear reaction sequence to afford the same molecule include better yields, ease of synthetic access to analogues, and a single purification step (Fig. 1 and Fig. 2).

Allenes are an underused functionality in organic synthesis, yet show great potential, as they can react with a vast array of functional groups [4]. The reaction of an aryl palladium(II) intermediate with an allene yields a highly reactive π-allyl palladium species which may then undergo an array of transformations. Examples of attack by nucleophiles [5], electrophiles [6], and transmetallation [7] have been reported. As part of our ongoing interest in palladium catalysed cascade chemistry, we herein report the synthesis of a small library of novel substituted (*Z*)-2-butene 1,4-diamine analogues and substituted 4-aryloxy but-2-enyl amine analogues, by nucleophilic attack on the π-allyl palladium species by an amine. A diverse range of amines and aryl iodides were chosen in order to investigate the effects of component sterics and electronic properties on reaction stereoselectivity and regioselectivity.

We started our investigation by synthesizing a range of allenes to be used in the palladium catalysed, three-component cascade reaction. A total of four allenes were prepared in good yields from *N*-propargyl phthalimides and *O*-propargyl phenols *via* the Crabbé reaction (Scheme 1).

The allenes were then used in the palladium catalysed cascade reaction with a wide array of commercially available amines and aryl iodides with different steric and electronic properties (Scheme 2).

The reactions of phthalimide based allenes (**1a** and **1b**) with a range of aryl iodides and amines were first assessed. Yields ranged from 31 to 90% and showed high stereoselectivity for the *Z*-isomer (Table 1). Aryl iodides with both electron-donating and electron-withdrawing groups are tolerated and had minimal effect on the yield, stereoselectivity, and regiospecificity. The products were obtained as a mixture of inseparable *E* and *Z* isomers. The preference for *Z* geometry was established by analogy with similar molecules [8–9]. The *E*/*Z* ratio was calculated by comparing the integrals of the alkenyl protons.

The reaction of phenol based allenes (**1c** and **1d**) with a range of aryl iodides was then assessed. Yields ranged from 31 to 67% with moderate selectivity for the *Z*-isomer (Table 2). The yields observed using the phenol based allenes were comparable to the phthalimide based allenes, however the reaction is less stereoselective.

The proposed reaction mechanism is outlined in Fig. 3. Pd(II) is first reduced to Pd(0) using PPh_3. The aryl iodide then undergoes oxidative insertion with Pd(0). Allene insertion then displaces one of the PPh_3 ligands from Pd(0), which is followed by migratory insertion of the aryl group. The complex can then undergo nucleophilic attack by the amine to liberate the product. The catalyst can be recovered through reductive elimination by a base [9].

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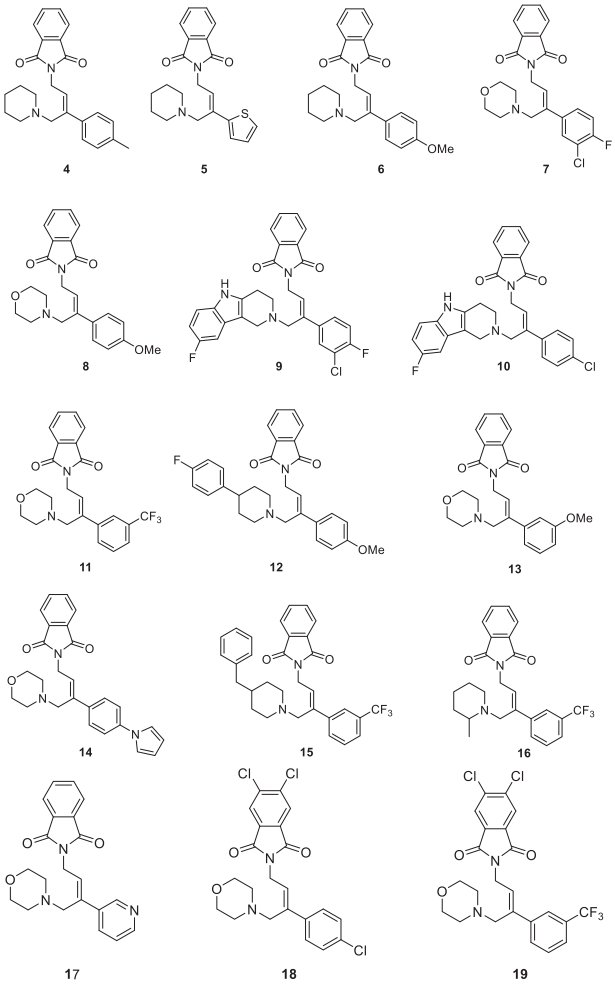


Fig. 1. Major isomers (*Z*) of the products formed from phthalimide based allenes in the palladium catalysed cascade reaction.

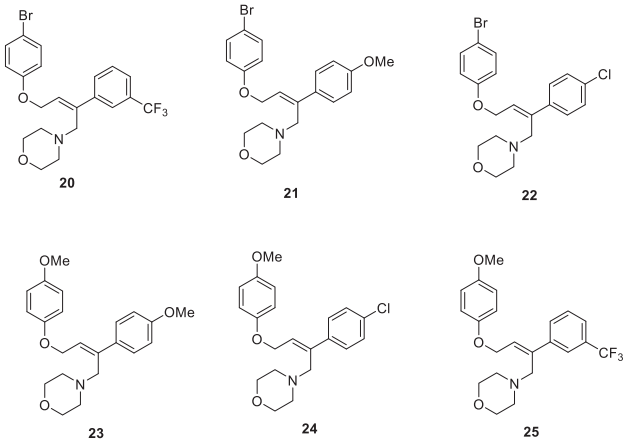
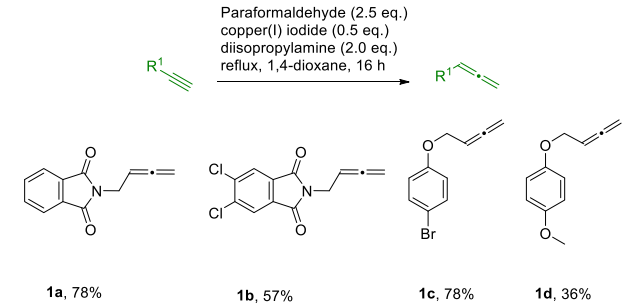
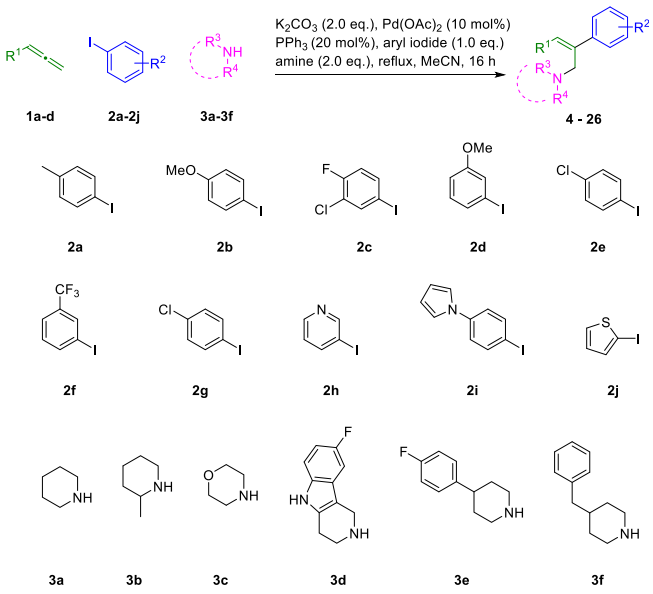


Fig. 2. Major isomers (*Z*) of the products formed from phenol based allenes in the palladium catalysed cascade reaction.

The regiospecific nature of the reaction can be explained by the equilibrium of the anti π-allyl complex with the two η^1 palladium complexes (Scheme 3). As expected, the nucleophile preferentially attacks the less sterically hindered complex to form the α-adduct [9].



Scheme 1. Synthesis of a range of allenes *via* the Crabbé reaction for use in the palladium catalysed cascade.



Scheme 2. The range of amines and aryl iodides used in the cascade reaction and the general reaction scheme.

Table 1
Combination of the aryl iodide, amine and allene used, along with the product obtained, yield and stereoselectivity.

Allene	Aryl Iodide	Amine	Product	Yield (%)	*E*/*Z*
1a	2a	3a	4	53	6:94
1a	2j	3a	5	78	0:100
1a	2b	3a	6	55	0:100
1a	2c	3c	7	76	17:83
1a	2b	3c	8	65	29:71
1a	2c	3d	9	73	22:78
1a	2e	3d	10	75	12:88
1a	2f	3c	11	74	21:79
1a	2b	3e	12	63	20:80
1a	2d	3c	13	90	16:84
1a	2i	3c	14	36	4:96
1a	2f	3f	15	31	10:90
1a	2f	3b	16	40	2:98
1a	2h	3c	17	36	0:100
1b	2g	3c	18	53	17:83
1b	2f	3c	19	61	12:88

If the γ-adduct was obtained, we would expect to see two germinal alkene protons in the ^{1}H NMR spectra. Instead, a single vicinal alkene proton appears as a triplet indicating that the α-adduct is exclusively formed.

Table 2
Palladium catalysed three component reaction.

Allene	Aryl Iodide	Amine	Product	Yield (%)	*E*/*Z*
1c	2f	3c	20	43	26:74
1c	2b	3c	21	53	29:71
1c	2 g	3c	22	36	25:75
1d	2b	3c	23	67	25:75
1d	2 g	3c	24	64	25:75
1d	2f	3c	25	61	30:70

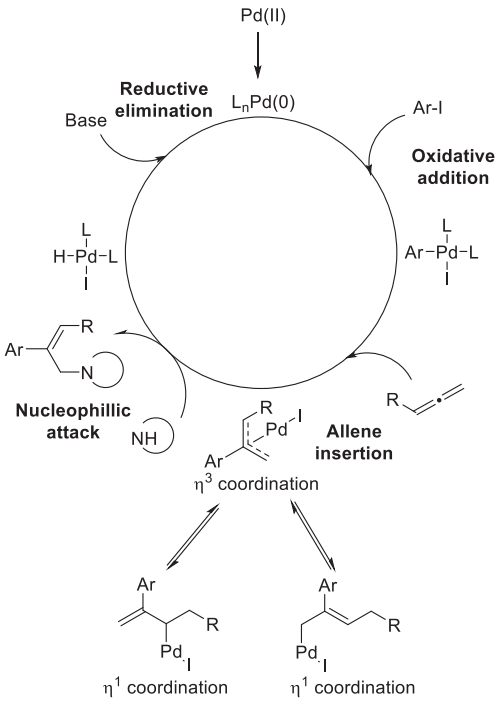
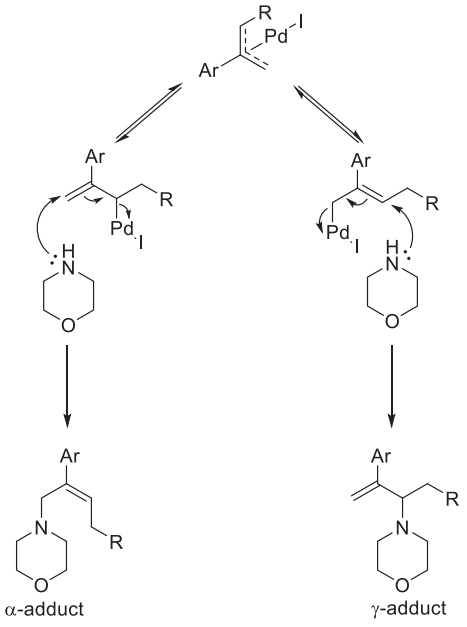
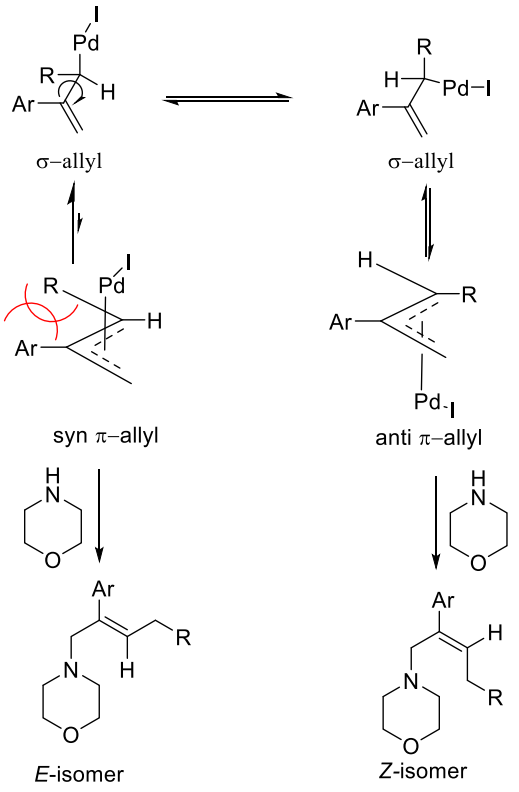


Fig. 3. Proposed reaction mechanism for the palladium catalysed, three-component cascade reaction.



Scheme 3. Regiochemical outcome of nucleophilic attack on both η^1 palladium complexes. Nucleophilic attack on the least hindered position is favoured (αadduct).



Scheme 4. Unfavourable interactions in the syn π-allyl complex leads to preferential formation of the anti π-allyl complex and the Z-isomer.

The two π-allyl palladium species can interconvert through rearrangement of the two σ-allyl species (Scheme 4). The amine attacks the least hindered carbon on each π-allyl palladium complex to give a mixture of *E*/*Z* isomers. The higher preference for the *Z*-isomer seen with the phthalimide based allenes can be explained by its larger steric bulk, resulting is a more significant clash between the allene R group and the aryl ring in the syn π-allyl complex This further favours the formation of the anti π-allyl complex and the resulting *Z*-isomer [9].

In summary, we have successfully prepared a wide range of substituted (*Z*)-2-butene 1,4-diamine analogues and substituted 4-aryloxy but-2-enyl amine analogues using a palladium catalysed, three-component cascade reaction, with good yields, regiospecificity and stereoselectivity for the *Z*-isomer. A range of aryl iodides were used, including those with electron-withdrawing and electron-donating groups. Amines of different steric bulk were also be used, with no noticeable effect on the yield, regiospecificity or stereoselectivity.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.tetlet.2020.152536.

References

[1] R. Lei, Y. Wu, S. Dong, K. Jia, S. Liu, W. Hu, J. Org. Chem. 82 (2017) 2862–2869.
[2] K. Li, L. Chen, Y.X. Fan, Y. Wei, S.J. Yan, J. Org. Chem. 84 (2019) 11971–11982.
[3] Q.X. Zi, C.L. Yang, K. Li, Q. Luo, J. Lin, S.J. Yan, J. Org. Chem. 85 (2020) 327–338.
[4] R. Zimmer, C.U. Dinesh, E. Nandanan, F.A. Khan, Chem. Rev. 100 (2000) 3067–3125.
[5] R. Grigg, I. Köppen, M. Rasparini, V. Sridharan, Chem. Commun. (2001) 964–965.
[6] X. Gai, R. Grigg, S. Collard, J.E. Muir, Chem. Commun. (2000) 1765–1766.
[7] U. Anwar, R. Grigg, M. Rasparini, V. Savic, V. Sridharan, Chem. Commun. (2000) 645–646.
[8] R. Grigg, E.E. Elboray, S. Akkarasamiyo, N. Chuanopparat, H.A. Dondas, H.H. Abbas-Temirek, C.W.G. Fishwick, M.F. Aly, B. Kongkathip, N. Kongkathip, Chem. Commun. 52 (2016) 164–166.
[9] K. Ross, R. Grigg, J. Assoc. Lab Autom. 9 (2004) 103–108.